

Neural Network

$$X = [[x_1], [x_2] \dots]$$

$$y = [0, 1, 2 \dots]$$

$$W = \begin{bmatrix} \text{no of parameters} \\ \vdots \\ \text{no of neurons} \end{bmatrix}$$

$$J = \frac{1}{m} \sum y_i \log(a)$$

$$a_1 = w^{[1]}_0 x + b_1$$

$$z_1 = \frac{1}{1 + e^{-(w_1 x + b_1)}}$$

$$a_2 = w^{[2]}_0 z_1 + b^{[2]}$$

$$z_2 = \frac{1}{1 + e^{-(w^{[2]}_0 z_1 + b^{[2]})}} = \frac{1}{1 + e^{-a_2}}$$

$$a_3 = w^{[3]}_0 z_2 + b^{[3]}$$

$$J = \frac{1}{m} \sum -y_i \log(a_3)$$

$$\frac{dJ}{da_3} = -\frac{1}{m} \sum \frac{y_i}{a_3}$$

$$\frac{da_3}{dz_2} = \frac{da_3}{dz} = w^{[3]}_0 \Rightarrow \frac{dJ}{dw^{[3]}_0} = \left(-\frac{1}{m} \sum \frac{y_i}{a_3} \right) \times w^{[3]}_0$$

$$\frac{da_3}{db^{[3]}} = w^{[3]} \cdot z_2 \Rightarrow \frac{dJ}{db^{[3]}} = w^{[3]} \cdot z_2 \left(-\frac{1}{m} \sum \frac{y_i}{a_3} \right)$$

$$\frac{dz_2}{da_3} = \frac{da_3}{dz_2} = w^{[3]}$$

$$\frac{dz_2}{da_2} = \frac{e^{a_2}}{(e^{a_2} + 1)^2}$$

$$\frac{da_2}{dw^{[2]}} = z_1 \Rightarrow \frac{dz_2}{dw^{[2]}} = z_1 \cdot \frac{e^{a_2}}{(e^{a_2} + 1)^2}$$

$$\frac{da_3}{dz_2} = z_1 \cdot \frac{e^{a_2}}{(e^{a_2} + 1)^2} \cdot w^{[3]}$$

$$\boxed{\frac{dJ}{dw^{[2]}} = z_1 \cdot \frac{e^{a_2}}{(e^{a_2} + 1)^2} \cdot w^{[3]} \cdot \left(-\frac{1}{m} \sum \frac{y_i}{a_3} \right)}$$

$$\frac{da_2}{db^{[2]}} = w^{[2]} \cdot z_1 \Rightarrow \frac{dz_2}{db^{[2]}} = w^{[2]} \cdot z_1 \cdot \frac{e^{a_2}}{(e^{a_2} + 1)^2}$$

$$\frac{da_3}{dz_2} = w^{[3]} \cdot w^{[2]} \cdot z_1 \cdot \frac{e^{a_2}}{(e^{a_2} + 1)^2}$$

$$\frac{dJ}{db^{[2]}} = w^{[3]} \cdot w^{[2]} \cdot z_1 \cdot \frac{e^{a_2}}{(e^{a_2} + 1)^2} \cdot \left(-\frac{1}{m} \sum \frac{y_i}{a_3} \right)$$

$$\frac{da_2}{dz_1} = w^{[2]}$$

$$\frac{dz_1}{da_1} = \frac{e^{a_1}}{(e^{a_1} + 1)^2}$$

$$\frac{dz_1}{dw^{[1]}} = w \quad \frac{da_1}{dw^{[1]}} = x$$

$$\frac{dz_1}{dw^{[1]}} = x \cdot \frac{e^{a_1}}{(e^{a_1} + 1)^2} \Rightarrow \frac{dz_1}{dw^{[1]}} = w^{[1]}$$

$$\frac{da_2}{dw^{[2]}} = w^{[2]} \cdot x \cdot \frac{e^{a_1}}{(e^{a_1} + 1)^2}$$

$$\frac{dz_2}{dw^{[2]}} = w^{[2]} \cdot x \cdot \frac{e^{a_1}}{(e^{a_1} + 1)^2} \cdot \frac{e^{a_2}}{(e^{a_2} + 1)^2}$$

$$\frac{da_3}{dw^{[3]}} = w^{[3]} \cdot w^{[2]} \cdot x \cdot \frac{e^{a_1}}{(e^{a_1} + 1)^2} \cdot \frac{e^{a_2}}{(e^{a_2} + 1)^2}$$

$$\frac{dJ}{dw^{[3]}} = w^{[3]} \cdot w^{[2]} \cdot x \cdot \frac{e^{a_1}}{(e^{a_1} + 1)^2} \cdot \frac{e^{a_2}}{(e^{a_2} + 1)^2} \cdot \left(-\frac{1}{m} \sum \frac{y_i}{a_3} \right)$$

$$\frac{da_1}{dw^{[1]}} = 1 \Rightarrow \frac{dz_1}{dw^{[1]}} = w^{[1]}$$

$$\frac{dJ}{dw^{[1]}} = w^{[1]} \quad \frac{da_1}{dw^{[1]}} = 1 \Rightarrow \frac{dz_1}{dw^{[1]}} = \frac{e^{a_1}}{(e^{a_1} + 1)^2}$$

$$\frac{da_2}{dw^{[2]}} = w^{[2]} \cdot \frac{e^{a_1}}{(e^{a_1} + 1)^2} \Rightarrow \frac{dz_2}{dw^{[2]}} \cdot \frac{dJ}{dw^{[1]}} = w^{[3]} \cdot w^{[2]} \cdot \frac{e^{a_1}}{(e^{a_1} + 1)^2} \cdot \frac{e^{a_2}}{(e^{a_2} + 1)^2} \cdot \left(-\frac{1}{m} \sum \frac{y_i}{a_3} \right)$$